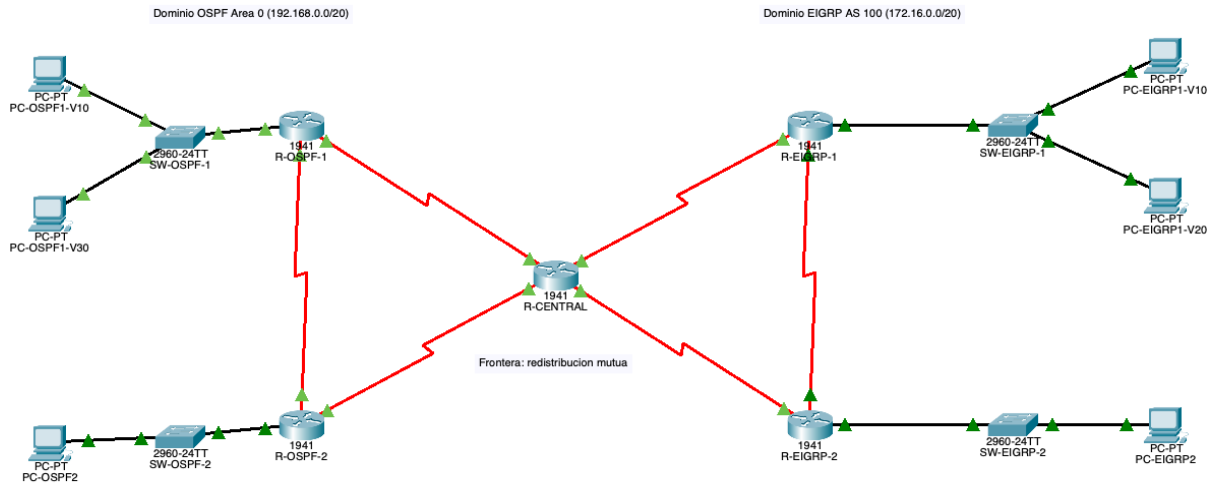
Laboratorio 3: Algoritmos de Enrutamiento

Evidencias de configuración y pruebas

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Topología implementada



1. Tablas de enrutamiento (show ip route)

Tomadas después de configurar OSPF, EIGRP y la redistribución en R-CENTRAL.

R-OSPF-1

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/128] via 10.0.0.6, 00:05:51, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.6, 00:04:14, Serial0/0/1
O    192.168.0.0/24 [110/129] via 10.0.0.2, 00:05:51, Serial0/0/0
                    [110/129] via 10.0.0.6, 00:05:51, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

R-OSPF-2

```
R-OSPF-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.10, 00:05:57, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.10, 00:04:14, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/129] via 10.0.0.1, 00:05:57, Serial0/0/0
                       [110/129] via 10.0.0.10, 00:05:57, Serial0/0/1
O       192.168.1.64/28 [110/129] via 10.0.0.1, 00:05:57, Serial0/0/0
                        [110/129] via 10.0.0.10, 00:05:57, Serial0/0/1
```

R-CENTRAL

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/192] via 10.0.0.5, 00:06:02, Serial0/0/0
                    [110/192] via 10.0.0.9, 00:06:02, Serial0/0/1
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/1/0
L       10.0.0.14/32 is directly connected, Serial0/1/0
C       10.0.0.16/30 is directly connected, Serial0/1/1
L       10.0.0.18/32 is directly connected, Serial0/1/1
D       10.0.0.20/30 [90/2707456] via 10.0.0.13, 00:06:10, Serial0/1/0
                     [90/2707456] via 10.0.0.17, 00:06:09, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/2170112] via 10.0.0.17, 00:06:09, Serial0/1/1
D       172.16.0.128/26 [90/2172416] via 10.0.0.13, 00:06:10, Serial0/1/0
D       172.16.0.192/27 [90/2172416] via 10.0.0.13, 00:06:10, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:06:02, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.9, 00:06:02, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.5, 00:06:02, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.5, 00:06:02, Serial0/0/0
```

R-EIGRP-1

```
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6585856] via 10.0.0.14, 00:06:05, Serial0/0/1
D EX    10.0.0.4/30 [170/6585856] via 10.0.0.14, 00:06:12, Serial0/0/1
D EX    10.0.0.8/30 [170/6585856] via 10.0.0.14, 00:06:12, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/1
L       10.0.0.13/32 is directly connected, Serial0/0/1
D       10.0.0.16/30 [90/6585856] via 10.0.0.14, 00:06:13, Serial0/0/1
                     [90/6611456] via 10.0.0.22, 00:06:12, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/0
L       10.0.0.21/32 is directly connected, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.16.0.0/25 [90/2195712] via 10.0.0.22, 00:06:14, Serial0/0/0
                      [90/6586112] via 10.0.0.14, 00:06:12, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D    192.168.0.0/23 [90/6585856] via 10.0.0.14, 00:06:05, Serial0/0/1
```

R-EIGRP-2

```
R-EIGRP-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6585856] via 10.0.0.18, 00:06:06, Serial0/0/1
D EX    10.0.0.4/30 [170/6585856] via 10.0.0.18, 00:06:11, Serial0/0/1
D EX    10.0.0.8/30 [170/6585856] via 10.0.0.18, 00:06:11, Serial0/0/1
D       10.0.0.12/30 [90/6585856] via 10.0.0.18, 00:06:11, Serial0/0/1
                     [90/6611456] via 10.0.0.21, 00:06:13, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
C       10.0.0.20/30 is directly connected, Serial0/0/0
L       10.0.0.22/32 is directly connected, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/2198016] via 10.0.0.21, 00:06:15, Serial0/0/0
                        [90/6588416] via 10.0.0.18, 00:06:11, Serial0/0/1
D       172.16.0.192/27 [90/2198016] via 10.0.0.21, 00:06:15, Serial0/0/0
                        [90/6588416] via 10.0.0.18, 00:06:11, Serial0/0/1
D    192.168.0.0/23 [90/6585856] via 10.0.0.18, 00:06:06, Serial0/0/1
```

2. Evidencia del balanceo OSPF (show ip ospf interface en R-OSPF-1)

Serial0/0/0 (enlace directo a R-OSPF-2) tiene costo 128 y Serial0/0/1 (hacia R-CENTRAL) tiene costo 64. Directo: 128 + 1 = 129. Por R-CENTRAL: 64 + 64 + 1 = 129. En la tabla de R-OSPF-1 la red 192.168.0.0/24 aparece con dos caminos [110/129], uno por 10.0.0.2 y otro por 10.0.0.6. En R-OSPF-2 pasa lo mismo con 192.168.1.0/26 y 192.168.1.64/28.

```
R-OSPF-1#show ip ospf interface

GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
```

3. Evidencia del balanceo EIGRP (show ip eigrp topology en R-EIGRP-1)

Para 172.16.0.0/25 (LAN de R-EIGRP-2) el sucesor es 10.0.0.22 por el enlace directo con métrica 2,195,712. El sucesor factible es 10.0.0.14 por R-CENTRAL con métrica 6,586,112 y RD 2,170,112, que es menor que la FD. Con variance 3 ambas rutas se instalan. La relación 6,586,112 / 2,195,712 = 3 reparte el tráfico 75 % por el enlace directo y 25 % por R-CENTRAL.

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)
```

```
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 6585856
         via Rstatic (6585856/2169856)
P 10.0.0.4/30, 1 successors, FD is 6585856
         via Rstatic (6585856/2169856)
P 10.0.0.8/30, 1 successors, FD is 6585856
         via Rstatic (6585856/2169856)
P 10.0.0.12/30, 1 successors, FD is 6073856
         via Connected, Serial0/0/1
P 10.0.0.16/30, 2 successors, FD is 6585856
         via 10.0.0.14 (6585856/2169856), Serial0/0/1
         via 10.0.0.22 (6611456/6073856), Serial0/0/0
P 10.0.0.20/30, 1 successors, FD is 2195456
         via Connected, Serial0/0/0
P 172.16.0.0/25, 2 successors, FD is 2195712
         via 10.0.0.22 (2195712/2816), Serial0/0/0
         via 10.0.0.14 (6586112/2170112), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 6585856
         via 10.0.0.14 (6585856/2169856), Serial0/0/1
```

4. Prueba de conectividad extremo a extremo

Ping y tracert desde PC-OSPF1-V10 (192.168.1.10, VLAN 10 de R-OSPF-1) hacia PC-EIGRP2 (172.16.0.10, LAN de R-EIGRP-2). El tracert pasa por R-OSPF-1 (192.168.1.1), R-CENTRAL (10.0.0.6) y R-EIGRP-2 (10.0.0.17).

